



Gas-liquid low-pressure forming of concave thin-walled aluminum alloy tubes

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Received: 26 May 2026 / Accepted: 23 September 2026

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Abstract

This study proposes a gas-liquid low-pressure forming (GLLF) process to address the wrinkling instability and severe local thinning encountered in manufacturing complex concave thin-walled metal tubes. Experiments and finite-element simulations were conducted on 6061 aluminum alloy tubes with concave angles of 30°, 45°, and 60°. The theoretical and experimental results revealed that the dynamic compressed gas-liquid mixture generates a nonlinear adaptive pressure path. This path exhibited an initial steady increase and a later exponential rise, which could safely meet the strict pressure requirements necessary to suppress the wrinkling instability during the compression of the tube. By elucidating the defect suppression mechanism within a “tension-bending-compression” mechanical framework, this process promotes a fundamental deformation shift from tensile thinning to compressive accumulation. Consequently, high wall thickness uniformity is achieved with maximum thinning rate kept within 5.5%, and the global three-dimensional profile deviation is strictly controlled within 0.5 mm. This study provides a reliable, high-precision and defect-free manufacturing method for complex concave tubes, providing an effective example for lightweight engineering in the aerospace and automotive industries.

Keywords Gas-liquid low-pressure forming · Concave thin-walled tube · Self-pressurizing mechanism · Wrinkling · Instability

1 Introduction

In recent years, lightweighting has emerged as a core development trend in modern advanced manufacturing, with significant application value in key fields such as aerospace and automotive engineering [1–2]. It not only reduces the load on power and transmission systems, improves the dynamic performance and handling stability of vehicles, but also effectively saves raw materials, reduces exhaust emissions, and contributes to achieving energy conservation and environmental protection goals [3]. Ma et al. [4] pointed out that lightweighting was one of the key strategies for the automotive industry to reduce energy consumption and emissions. At present, the main approaches to

achieving lightweighting include material lightweighting [5] and structural lightweighting [6]. Material lightweighting mainly relies on the application of low-density alloys such as aluminum and magnesium [7–8], while structural lightweighting mainly improves the performance-to-weight ratio of components by optimizing the geometric shape of tubular parts. Hollow variable-section tubes have attracted increasing attention due to their dual advantages in both material and structural design. Lang et al. [9] systematically summarized the advantages of such tubes in significantly reducing mass while maintaining structural strength under dominant bending and torsional loads. Among these, concave thin-walled metal tubes exhibit broad application prospects in lightweight structural design and manufacturing due to their high specific strength, excellent bending and torsional resistance, and good assembly stability. Lin et al. [10] conducted in-depth research on the instability mechanisms of the positive-curvature arc regions of such tubes during the forming process, further promoting their engineering applications.

Currently, a variety of manufacturing technologies are available for concave thin-walled metal tubes. Although

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various technologies such as stamping-welding, roll forming, and additive manufacturing are available for producing concave tubes, their engineering applications are often constrained by weld-induced fatigue weakness, low production efficiency, or material anisotropy [11–13]. Han et al. [14] investigated the use of custom laminated tubes for hydroforming to produce hollow tubular components with variable thickness without welding. Talebi-Anarakhi et al. [15] studied the thermal gas forming process of aluminum alloy tubes using a heating method, which effectively reduced the deformation resistance of the material at high temperatures. Although additive manufacturing and roll forming have their respective advantages, hydroforming remains the mainstream process for manufacturing complex concave thin-walled metal tubes due to its high forming precision and excellent lightweighting effect.

The driving mechanism of hydroforming originates either from the active bulging of the internal medium (high-pressure) or mechanical compression by the external die (low-pressure) [16]. Traditional high-pressure forming relies on ultra-high internal pressure (up to 400 MPa) or elevated temperatures to expand the tube blank, which significantly improves component stiffness but often leads to severe corner thinning, high equipment tonnage, and elevated energy consumption [17–18]. To reduce forming pressure and improve wall thickness distribution, low-pressure forming processes (e.g. tube hydro-forming and liquid-filled radial upseting) utilize die closure as the main driving force while internal fluid provides adaptive support, lowering required pressure by 30%–50% [19–22]. However, Ma et al. [23] noted that concave thin-walled tubes remain highly prone to wrinkling during compression. Furthermore, because

conventional low-pressure forming uses incompressible liquid media, minor internal volume changes during die clamping induce severe pressure fluctuations, imposing strict requirements on pressure control and sealing [20, 24].

To address the pressure control challenges associated with liquid media and combine the merits of both high-pressure and low-pressure forming, Yuan et al. [25] proposed the Gas-Liquid Low-pressure Forming (GLLF) process for manufacturing square tubes, which can also be extended to produce concave tubes. Furthermore, the relevant studies have provided more reliable support via finite element simulations and novel process explorations for wrinkling control and complex deformation boundary prediction during forming [26–29]. This enables the GLLF process to effectively integrate the synergistic effects of die compression and internal adaptive pressure support. Recent research progress demonstrates that it offers significant advantages in overcoming the forming limits associated with small corner radii [30].

To intuitively illustrate these advantages, a comparative evaluation of typical forming processes for complex concave thin-walled metal tubes is presented in Fig. 1. Hot gas forming is representative of high-pressure forming. It is primarily characterized by bulging accompanied by significant material thinning, which often results in poor dimensional control of the final component. In the traditional low-pressure forming process, the use of incompressible liquid media often leads to severe pressure fluctuations when the mold is closed. This causes the tube wall to be highly prone to bending defects and imposes extremely high requirements on the sealing performance of the equipment. In contrast, the gas-liquid low-pressure forming (GLLF) process utilizes

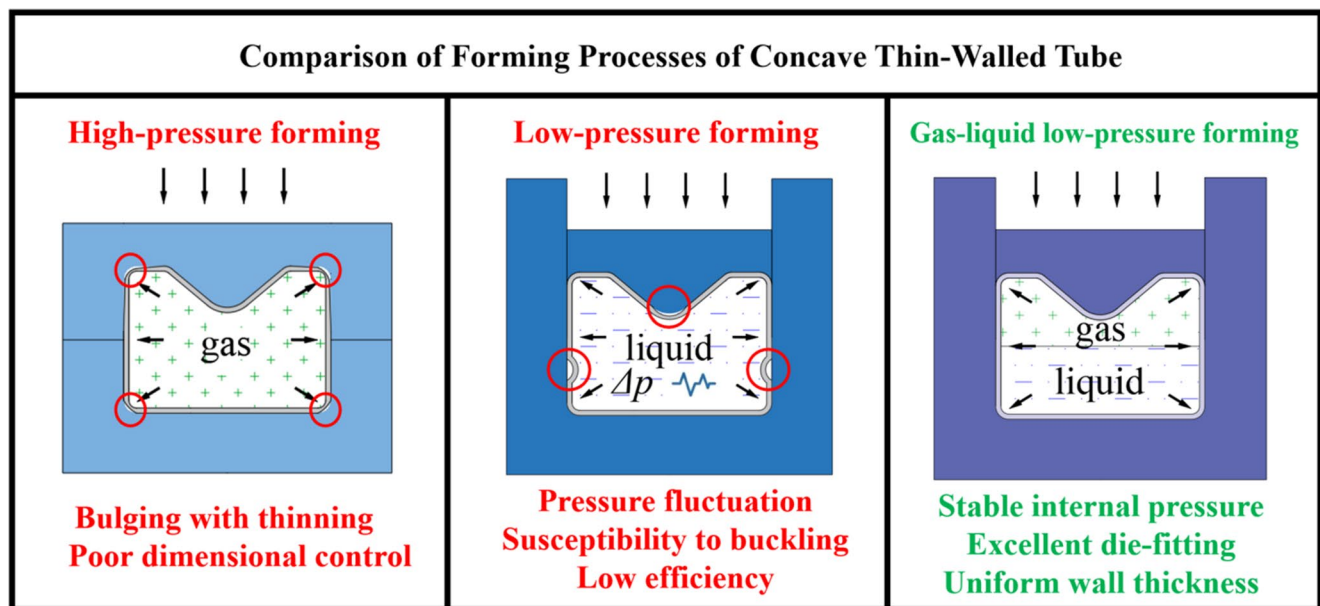


Fig. 1 Comparison of typical forming processes for complex concave thin-walled metal tubes

a highly compressible gaseous phase to provide an adaptive pressure buffering mechanism. This mechanism can smoothly compensate for the changes in the internal volume of the tube component, thereby achieving a stable internal pressure environment. Therefore, it helps to achieve good die fitting and maintain a more uniform wall thickness on the concave contour, successfully overcoming the inherent limitations of single-medium forming techniques.

In this study, the Gas-Liquid Low-pressure Forming (GLLF) process was employed to fabricate complex concave thin-walled metal tubes. The different gas-liquid volume ratios were set to enable the internal pressure of the tubes to adaptively adjust according to volume changes by taking advantage of the high compressibility of gases, thereby avoiding sudden pressure surges and improving the forming stability. The main contributions are as follows:

- (1) A gas-liquid mixed medium compression model was established to quantitatively predict internal pressure evolution and dynamic loading paths.
- (2) Anti-buckling and die-fitting criteria were derived and the required pressure to suppress wrinkling defects was figured out to suppress wrinkling defects.
- (3) The proposed model was validated across various concave angles through experiments and FE method, clarifying the deformation mechanism and material flow behavior.

2 Principle of gas-liquid low-pressure forming

The gas-liquid low-pressure forming (GLLF) process uses the synergistic effect between the compressible gas phase and the nearly incompressible liquid medium to achieve a stable and adaptive internal pressure environment during the mechanical compression of thin-walled tubes. As shown in the progressive forming sequence diagram in Fig. 2(a) and the corresponding displacement-pressure evolution path in Fig. 2(b), the entire process can be systematically divided into four consecutive stages: the proceeding from the initial positioning to the final high-precision compression.

In the initial stage, the circular tube billet is first positioned in the lower die cavity, and then the upper die moves downward until it establishes the initial physical contact with the outer surface of the tube (Stage I: Pre-closure).

Subsequently, a predetermined volume of incompressible liquid is injected, followed by gas charging to achieve the preset initial pressure P_0 , thereby forming the internal gas-liquid mixture (Stage II: Medium filling). Upon completion of end sealing, the initial gas-liquid volume ratio k becomes

the dominant factor governing the subsequent nonlinear hardening behavior and pressure evolution of the mixture.

As the upper die continues to move downward, the process enters the profile forming stage (Stage III), where the main deformation mode is large-scale bending, forcing the circular cross-section to conform to the macroscopic concave contour of the die. The internal volume of the tube decreases moderately during this stage, leading to a relatively smooth and gradual rise in internal pressure. The self-adaptive buffering capacity provided by the high compressibility of the gas effectively absorbs the volume changes and avoids the common sharp pressure peaks in traditional incompressible fluid hydraulic forming, thereby preventing early tube burst or excessive outward expansion.

The final and most critical stage is Stage IV (Section compression), in which the die approaches complete closure, and the straight sections of the tube undergo intense in-plane biaxial compression. The formation of the final deep concave structure results in a substantial reduction in the internal volume of the tube, causing the internal pressure P to increase exponentially. This “self-pressurization” mechanism generates a strong, dynamically enhanced normal supporting force that counteracts the bending tendency of the sidewalls. The GLLF process suppresses the instability of wrinkles by perfectly matching the strict pressure requirements of the later compression stage and can achieve high-precision manufacturing of complex concave contours under the “stretch-compression-bend” composite mechanical mechanism.

3 Theoretical modeling

The geometrically distinct regions of the tube cross-section—convex corners, concave corners and straight-edge sections—exhibit markedly different mechanical states and deformation behaviors during the gas-liquid low-pressure forming process. Based on the principles of plasticity mechanics and deformation energy conservation, and drawing upon established research on tube instability failure criteria, this paper presents an in-depth derivation of the force mechanisms and the critical conditions governing die adhesion and instability in each region, thereby enabling accurate prediction of the internal pressure required for forming while preventing local instability. The deformation model of the cross-section forming is illustrated in Fig. 3(a). The deformation behavior of the corner and straight-edge regions can be analyzed using plasticity mechanics and the energy method. According to their deformation characteristics, the corners are categorized into convex corners (outward bending) and concave corners (inward bending), as depicted in

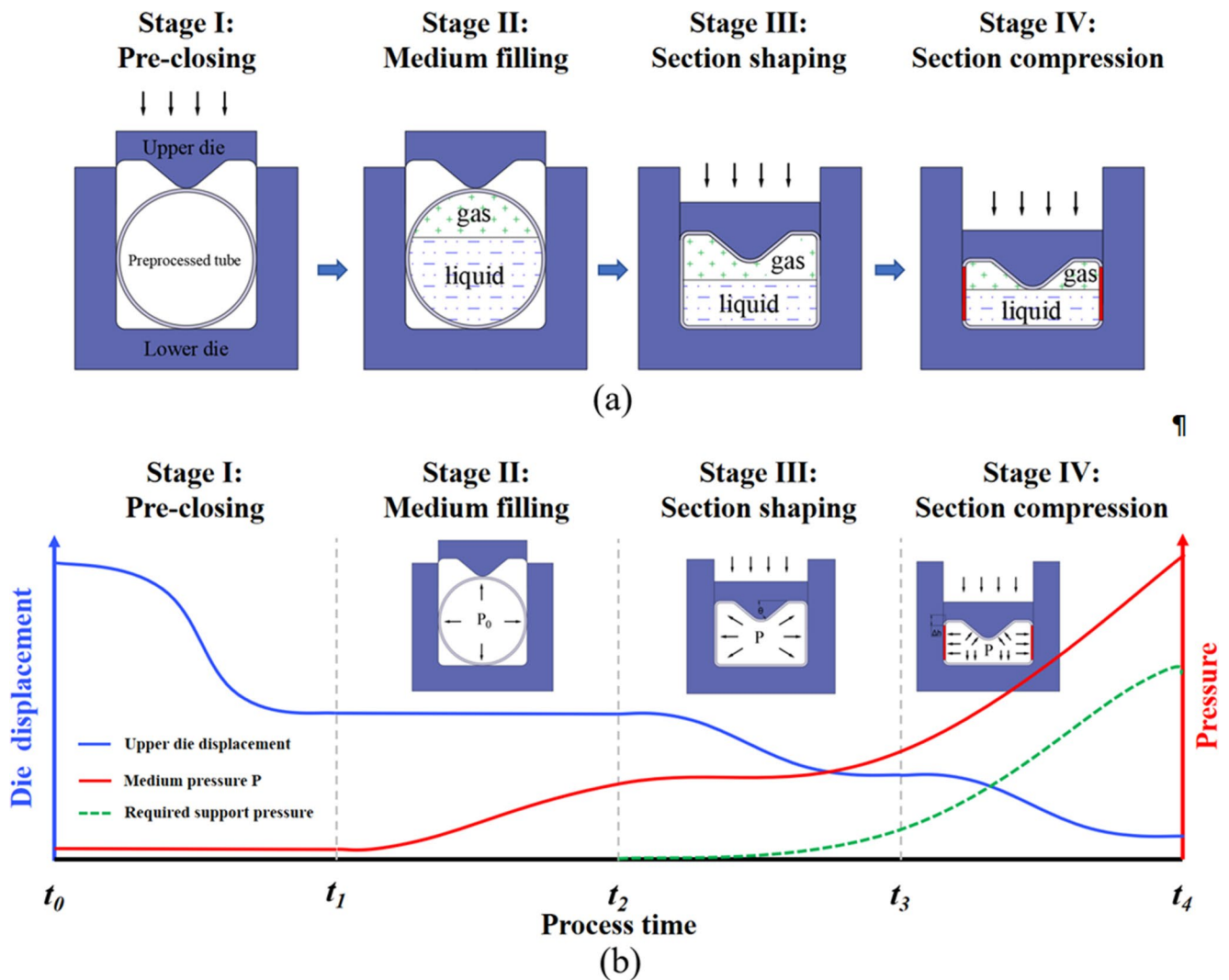


Fig. 2 The principle and typical deformation stages of the gas-liquid low-pressure forming (GLLF) process: **(a)** schematic diagram of the progressive forming sequence from pre-closing to section compression; **(b)** displacement-pressure evolution path and the corresponding self-pressurizing mechanism

Fig. 3(b) and (c), respectively. These two corner types differ substantially in terms of their stress states and die adhesion behavior.

3.1 Pressure requirements for convex-corner die fitting

To guide the selection of experimental parameters (such as initial pressure and gas fraction) and to predict the instability boundaries of concave tubes, a theoretical model based on plasticity mechanics and the energy method is established as follows.

The following simplifying assumptions are adopted: the tube is treated as a thin-walled shell subjected to plane strain conditions; the 6061 aluminum alloy is assumed to be isotropic and obeys a power-law hardening model.

As depicted in Fig. 3(b), the convex corner region primarily undergoes outward bending and die adhesion deformation under the combined action of die compression and internal fluid support. Lin et al. [10] established the mechanical model for the positive-curvature arc parts in tube hydroforming. The bending angle of the convex corner is θ , the bending radius is R , the initial wall thickness of the tube is t , and the material yield strength is σ_s . The clamping force F_d exerted by the die generates a local bending moment in the corner region. According to statics equilibrium, the bending moment per unit length of the cross-section M can be approximately expressed as Eq. (1).

$$M = \frac{F_d \cdot R \cdot \theta}{2} \quad (1)$$

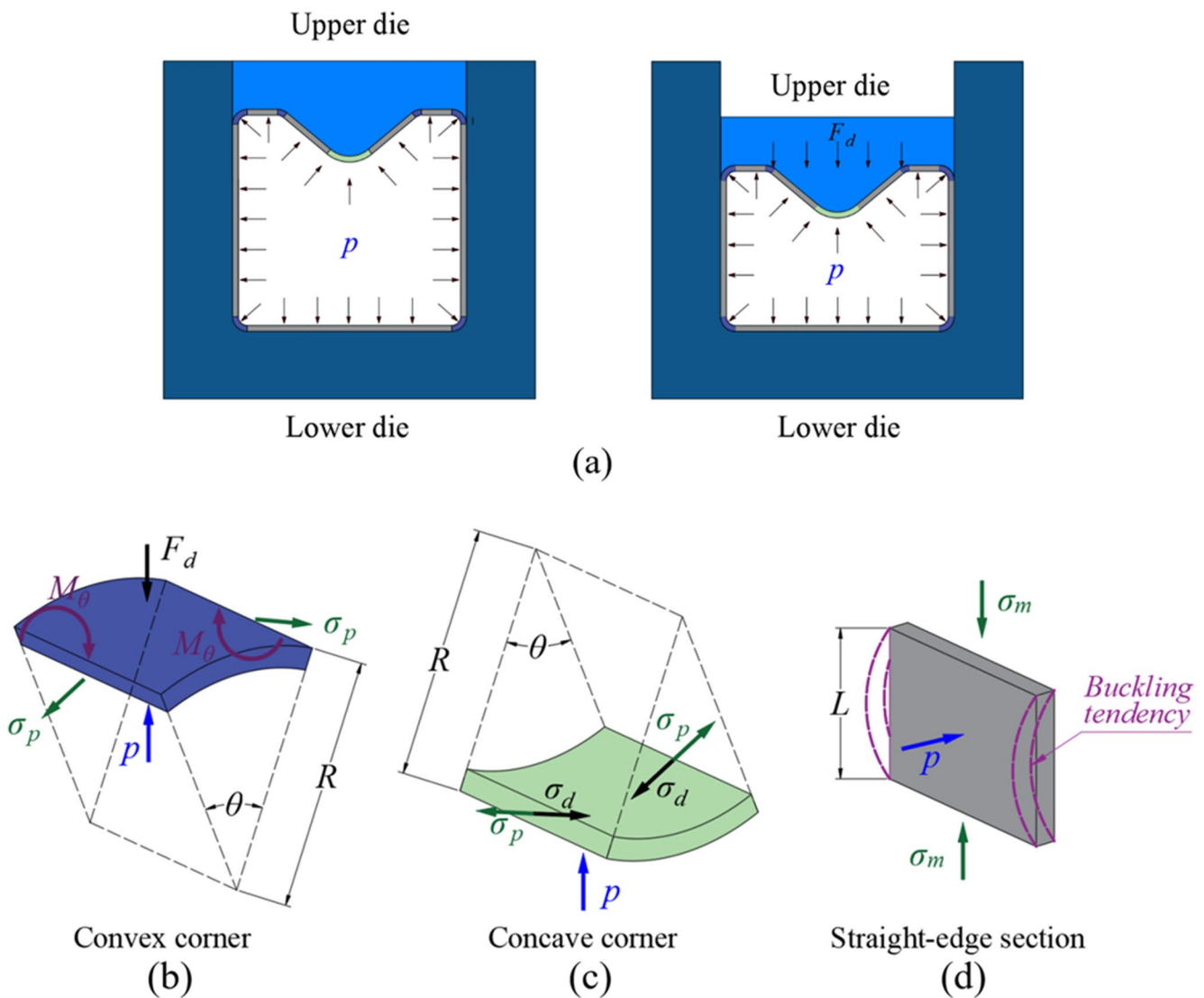


Fig. 3 Macro-sectional geometry and representative micro-element stress analysis of different regions during the gas-liquid hybrid compression forming process: **(a)** Overall macroscopic cross-section of the profile; **(b)** Schematic stress state of a representative micro-element in

the convex region; **(c)** Schematic stress state of a representative micro-element in the concave region; **(d)** Structural mechanics model and wrinkling instability criterion for the straight-edge panel

Simultaneously, the supporting pressure p exerted by the internal gas-liquid hybrid medium generates a circumferential tensile stress σ_p within the tube wall. According to the theory of thin-walled cylinders subject to internal pressure [31], this can be expressed as Eq. (2).

$$\sigma_p = \frac{p \cdot R}{t} \quad (2)$$

For the convex corner to fully adhere to the die, the work done by the external loads (including the work done by the bending moment and the tensile work caused by internal pressure) must overcome the plastic deformation energy W_p required for material yielding and flow. Based on the

classical energy balance equation in metal plastic forming, the mechanical criterion for die adhesion can be established as Eq. (3) [32].

$$\frac{M}{2EI} + \sigma_p \cdot \varepsilon_p \geq W_p \quad (3)$$

where E is the elastic modulus of the material, I is the cross-sectional moment of inertia, and ε_p is the equivalent plastic strain. By substituting Eqs. (1) and (2) into Eq. (3) and simplifying based on the material's yield criterion, the critical supporting pressure p_c required for the convex corner to achieve tight die adhesion can be derived as Eq. (4).

$$p_c \geq \frac{2\sigma_s t}{R} - \frac{F_\theta \cdot \theta}{2\pi R} \quad (4)$$

3.2 Pressure requirements for concave-corner die fitting

Contrary to convex corners, concave corners bend inward during the forming process, as schematically shown in Fig. 3(c). Let the concave corner angle be θ and the bending radius be R_i . In the early stage of deformation, a circumferential tensile stress exists at the concave corner due to the slight overall expansion of the tube, facilitating die adhesion. However, as the die closing compression advances, the strong extrusion from the die causes the circumferential stress in this region to rapidly reverse, turning into an extremely high compressive stress. Referring to the analytical model of die closing force and contact stress by Chu et al. [24], let the circumferential compressive stress induced by die compression be σ_p , and the circumferential tensile stress induced by internal pressure be σ_d . The actual circumferential stress σ_θ at the concave corner is the linear superposition of the two. If the net compressive stress is too large, the material will undergo reverse instability and detach from the die. The critical stress condition to prevent detachment defects is Eq. (5).

$$\sigma_\theta = \sigma_p - \sigma_d < \sigma_{cr} \quad (5)$$

where σ_{cr} is the critical buckling stress at which local instability occurs. By integrating the stress surface of the concave corner using the plastic deformation work method, the minimum internal pressure $p_{c,i}$ required to maintain the concave corner without detachment from the die can be obtained as Eq. (6).

$$p_{c,i} = \frac{K \cdot \sigma_s \cdot t}{R_i} \left(1 + \frac{\theta}{\pi}\right) \quad (6)$$

where K is a dimensionless coefficient related to the material's hardening characteristics. This formula indicates the limitations imposed by cross-sectional geometric features on the forming boundaries: the deeper the concave angle feature (i.e., the larger θ is) and the smaller the local radius R_i , the supporting internal pressure $p_{c,i}$ required to maintain die adhesion shows a significant nonlinear increase. This essentially explains why, in traditional low-pressure forming, the sharp concave features are most prone to non-adhesion and collapse defects.

3.3 Pressure requirements for straight-wall anti-buckling

During the compressive deformation stage of the cross-section, the straight-edge regions on both sides of the tube

serve as the primary load-bearing areas. As depicted in Fig. 3(d), their stress state resembles that of a thin elastic plate under in-plane biaxial compression. If there is not sufficient internal pressure support, macroscopic wrinkling instability (sidewall buckling) is highly likely to occur due to excessive compressive stress.

To explicitly establish the wrinkling suppression criterion during the section compression stage, a simplified mechanical model based on classical thin-plate buckling theory is introduced, as schematically illustrated in Fig. 3(a), (d). During die closure, the straight-edge sidewall is subjected to an in-plane circumferential compressive stress σ_m induced by the closing force F_d , while the internal pressure p provides supporting resistance against inward localized deformation. Here, the key theoretical parameters in the instability criterion are rigorously defined as follows:

L is the macroscopic effective length of the unsupported straight section in Fig. 3(d). The in-plane compressive stress σ_m acting on the straight section is derived from the upper die closing load F_d and profile geometry, which can be expressed as Eq. (7).

$$\sigma_m = \frac{F_d \cos(\frac{\theta}{2})}{2t \cdot L} \quad (7)$$

Where θ is the cavity corner angle, t is the nominal wall thickness of the tube, and L is the axial contact length of the die section.

According to classical elastic thin plate stability theory, the critical compressive stress σ_{buck} for buckling of the straight-edge section can be expressed as Eq. (8) [32].

$$\sigma_{buck} = \frac{\pi^2 E}{12(1 - \nu^2)} \left(\frac{t}{L}\right)^2 \quad (8)$$

In the GLLF process, the internal fluid pressure p generates normal support and circumferential tensile stress within the tube wall of the straight-edge section (approximately $P \cdot \frac{D}{2t}$, where D is the outer diameter of the tube), which will resist inward buckling. Therefore, the absolute mechanical boundary condition to ensure that the straight-edge section does not wrinkle is the macroscopic compressive stress induced by die downward displacement, minus the flexible supporting force provided by internal pressure. The difference between the two must not exceed the critical buckling stress limit of the thin plate, as shown in Eq. (9).

$$\sigma_d - \frac{p \cdot D}{2t} < \sigma_{buck} \quad (9)$$

By transforming Eq. (8), the dynamic critical supporting internal pressure $p_{cr, edge}$ required to prevent buckling of the straight-edge section can be obtained as Eq. (10).

$$p_{cr, edge} = \frac{2t}{D} (\sigma_d - \sigma_{buck}) \quad (10)$$

Equation (9) clearly indicates that because compressive stress increases sharply with the increase in die downward displacement, the critical internal pressure $p_{cr, edge}$ required to prevent sidewall wrinkling is also a dynamically climbing process at an exponential rate. This derivation theoretically confirms the nonlinear hardening characteristics of the mixed fluid medium in the GLLF process proposed by Yuan et al. [30], namely “as the compression volume decreases, it will achieve rapid self-pressurization”, which is in perfect agreement with the strict pressure requirements for the straight section’s resistance to bending.

The foregoing theoretical derivations clearly demonstrate that different geometric regions exhibit distinct instability tendencies during the gas–liquid low-pressure forming process. The straight-edge section is most susceptible to instability, as it undergoes a biaxial compressive stress state during the compression phase—analogous to a compressed thin plate—rendering it vulnerable to sidewall buckling. At the later stage of deformation, the circumferential stress at the concave corner transitions from tension to compression, causing the tube to separate from the die and resulting in defects such as die non-adhesion and compromised dimensional stability. The convex corner is the relatively most stable. Its circumferential compressive stress, in conjunction with the internal pressure, promotes die fitting and wall thickening, thereby reducing its susceptibility to instability.

3.4 Compression model for the gas–liquid mixed medium

The pressure within the tube changes constantly as the gas–liquid mixed medium is compressed. The relationship between the pressure and the compression of the medium is derived below. Assume that the gas follows an adiabatic process and the liquid has weak compressibility (bulk modulus K_l). The physical parameters of the model are shown in Table 1.

According to Poisson’s equation, the relationship between gas volume V_g and pressure is expressed as Eq. (10) [33].

$$V_g = kV_0 \left(\frac{P_0}{P} \right)^{\frac{1}{\gamma}} \quad (10)$$

Table 1 Physical parameters of the model

Symbol	Physical meaning	Unit
P_0	Initial pressure	MPa
P	Current pressure	MPa
V_0	Initial total volume	m ³
V	Current total volume	m ³
k	Initial gas volume fraction	-
λ	Volume compression ratio	-
K_l	Bulk modulus of liquid	Pa

The adiabatic assumption in Eq. (10) is adopted because the rapid compression in the GLLF process substantially limits the time for heat exchange between the gas, liquid, and die. Thus, thermal energy is largely retained within the gas, contributing to the observed self-pressurizing effect.

Simultaneously, based on the definition of liquid isothermal compressibility in fluid mechanics, the current liquid volume V_l is given by Eq. (11) [25].

$$V_l = (1-k)V_0 \left[1 - \frac{P - P_0}{K_l} \right] \quad (11)$$

Based on volume constraints and the definition of the compression ratio, the implicit governing equation relating pressure P and the compression ratio λ is derived as Eq. (12).

$$\lambda = 1 - kV_0 \left(\frac{P_0}{P} \right)^{\frac{1}{\gamma}} - (1-k) \left[1 - \frac{P - P_0}{K_l} \right] \quad (12)$$

This model provides a theoretical basis for pressure control in the GLLF process of concave thin-walled metal tubes. A stable pressure evolution curve can be achieved by rationally setting the initial gas fraction and compression path and effectively suppress the compression instability in thin-walled tubes. From this implicit equation, it can be inferred that the larger the value of k is, the steeper the P – λ curve will be in the later stage.

4 Materials and methods

4.1 Material

This study utilized 6061 aluminum alloy tubes initially in the T5 temper. The tubes were pre-annealed (held at 415 °C for 3 h) to convert them to the O temper to reduce the deformation resistance during the forming process. The tubes had an outer diameter of 38 mm, a wall thickness of 1.5 mm, and a length of 360 mm. At room temperature (25 °C), the density of the tube material was approximately 2.70 g/cm³, the elastic modulus was about 69 GPa, and the Poisson’s ratio was 0.33. The annealing treatment made the material softer, which significantly reduced its yield strength and ultimate tensile strength (from 224 MPa and 276 MPa to approximately 55 MPa and 125 MPa respectively).

4.2 Experimental setup

The experimental setup for the gas–liquid low-pressure forming (GLLF) process is shown in Fig. 4. The system adopted a highly integrated modular architecture to

achieve the simultaneous filling of multiple media and mechanical compression. The central forming module consisted of a press equipped with upper and lower mold components, where the tubular blank was placed and laterally sealed fixed by a dedicated sealing plug. The internal mixed media were established through two independent conveying paths: the gas filling module supplied gas through the left-side sealing plug from a high-pressure gas cylinder, while the liquid filling module injected the liquid medium through the right plug. The entire forming process was coordinated and completed by a centralized control module, which was equipped with a dedicated control console for executing operational logic and coordinating the mold stroke with the internal pressure. Additionally, an auxiliary hardware module was integrated to continuously collect data and monitor the dynamic response of the system. The coordinated operation of these functional units enabled the successful deformation of the blank, ultimately forming the complex concave contours shown in the frontal and vertical views of the target component.

The outer diameter of the circular tube used in this compression experiment was 38 mm, and its length was 360 mm. The initial perimeter was approximately 119.38 mm, and the initial internal volume was approximately 0.4 L. The punch gaskets with 30°, 45° and 60°

concave angles were used during the test. When the tube perimeter was not compressed, the maximum die downward displacements corresponding to the 30°, 45°, and 60° inserts were approximately 24 mm, 25 mm, and 26 mm, respectively. The final limit displacement was uniformly set to 27 mm. The experiments were conducted with different initial gas volume fractions (0.6, 0.7) and initial pressures (0.8–2.0 Mpa) corresponding to the 0 to 27 mm stroke. Specific parameters are listed in Table 2.

4.3 FE modeling setup

Numerical simulations of the GLLF process were carried out in ABAQUS according to the finite element model shown in Fig. 5(b). The aluminum alloy tube (O-temper) was discretized using 4-node reduced-integration shell elements (S4R), with 5 integration points through the thickness direction to accurately capture localized deformation. Isotropic plastic behavior was governed by the

Swift strain-hardening law ($\sigma = K(\epsilon_0 + \bar{\epsilon}_p)^n$). Interfacial contact was modeled via a surface-to-surface penalty formulation, enforcing “Hard Contact” in the normal direction and Coulomb friction ($\mu = 0.1$) in the tangential direction. Regarding boundary conditions, the lower die was completely fixed, while a vertical displacement

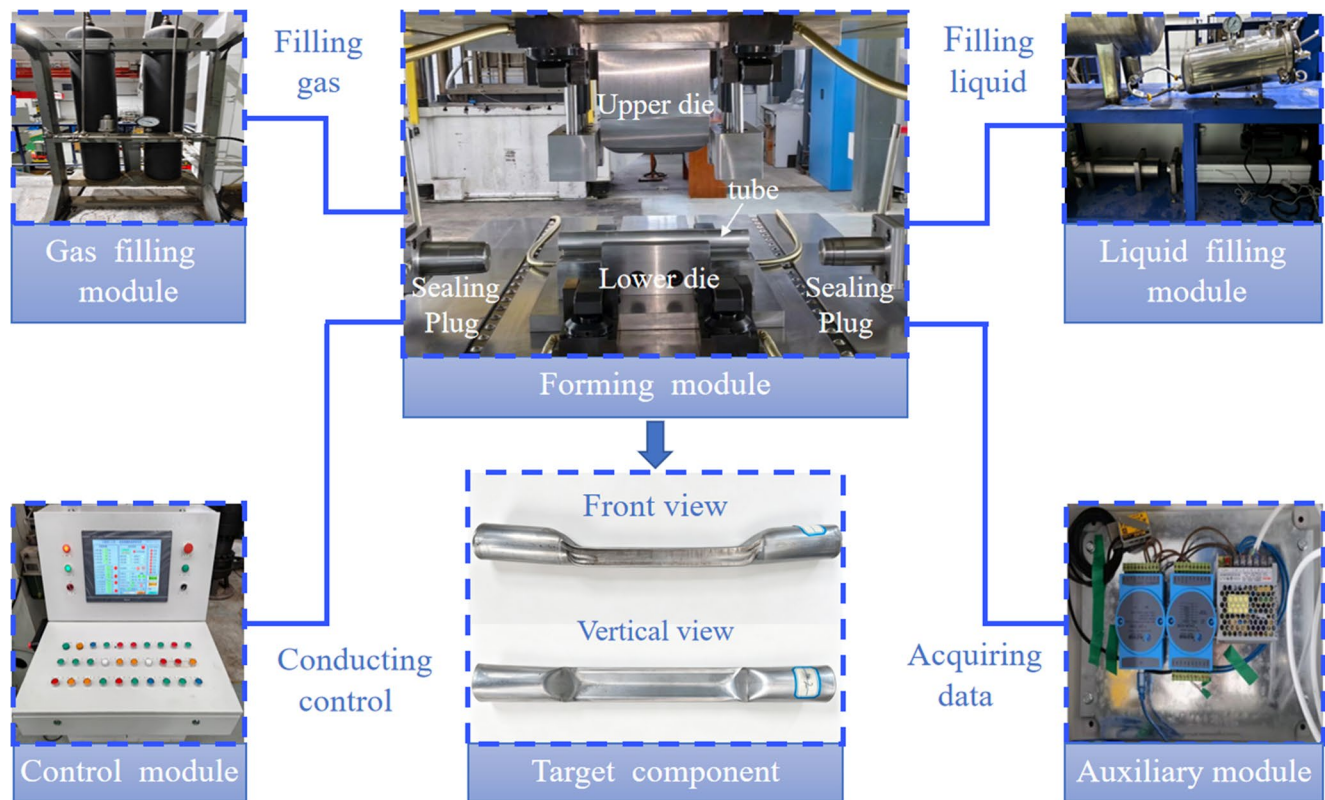


Fig. 4 Experimental system and key functional modules for the gas-liquid low-pressure forming (GLLF) process

Table 2 Experimental design matrix for concave tube forming

Concave angle (°)	Die stroke (mm)	Initial pressure (MPa)	Initial gas volume fraction
30°	0–27	0.8–2.0	0.6
45°			0.7
60°			0.7

was prescribed to the upper die. Rigid axial sealing constraints were simplified by fully restricting all degrees of freedom ($U = U_R = 0$) at Reference Points (RPs) coupled to both extremities of the tube.

To accommodate the adaptive internal support characteristic of GLLF, a volume-pressure-deformation coupling model was established (Fig. 5a). Because continuous cross-sectional deformation alters the internal cavity volume during die closure, a custom Python script was implemented in ABAQUS for step-by-step iterative calculations. At each solution increment, the script

dynamically calculated instantaneous volume variations, updated the internal medium pressure, and re-applied it to the inner tube wall, thereby achieving a closed-loop coupling between internal pressure evolution and structural deformation.

5 Results and discussion

5.1 Pressure for tube forming with different concave angles

As the die continues to move downward, the tube cross-section gradually evolves from an initial circular tube into a profiled cross-section with varying concave features.

To facilitate the subsequent analysis and discussion, the key process variables and geometric parameters of the concave tube forming process are defined in Table 3.

Fig. 5 FE modeling framework for GLLF: (a) volume-pressure-deformation coupling mechanism; (b) ABAQUS finite element model of the tube GLLF

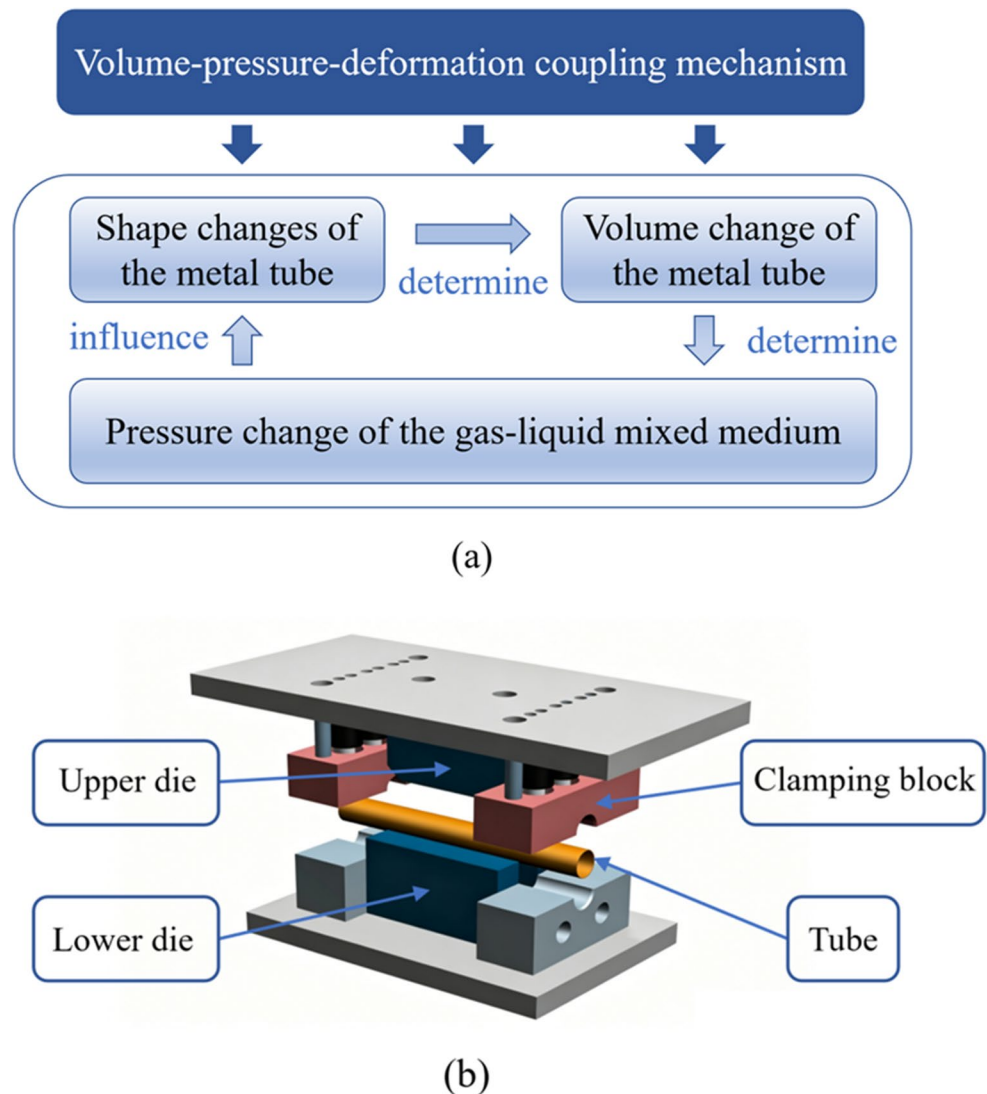


Table 3 Geometric parameters of the concave tube

Symbol	Physical meaning	Unit
L	Downward stroke of the upper die	mm
H	Height of the concave tube profile	mm
θ	Concave angle of the tube	°
t	Initial wall thickness	mm

As shown in Fig. 6(a), the geometric transition of the 30° concave tube is relatively gradual. When the downward displacement L reaches 24 mm (die closing height $H=19.9$ mm), it first completes the initial die fitting and enters the overall cross-sectional compression stage. As the downward displacement increases, wrinkles of varying degrees appear in the straight-edge section. This is mainly attributed to the continuous downward pressure of the upper die, which resulted in a certain degree of material accumulation at the circle corners areas of the tube bottom where the material flowability is relatively poor. When the concave angle increases to 45° (Fig. 6(b)), the onset of section compression is delayed to $L=25$ mm ($H=18.1$ mm) because of the deeper die penetration. At this moment, the more material participates in forming the concave angle, thereby reducing the perimeter of the straight-edge section. As shown in Fig. 6(c), the steepest 60° concave tube requires a longer forming stroke to fully shape the V-shaped feature. It is not until L reached 26 mm ($H=16.8$ mm) that the section entered the stage of intense compression stage. Moreover, the straight-edge section at the bottom of the tube only exhibits slight wrinkling at a downward displacement of $L=27$ mm.

Figure 7 presents the displacement-pressure evolution relationships and the corresponding forming process diagrams obtained from experimental tests on tubes with different concave angles. These experimental trajectories directly validate the gas-liquid hybrid medium compression model and the anti-buckling criteria established in Sect. 3.4. Taking the 30° concave tube as an example (Fig. 7(a)), the internal mixed medium pressure exhibits a significant non-linear climbing trend with the die downward displacement (L) at different initial pressures. As shown in the forming process diagram (Fig. 7(b)), the critical support pressure curve for wrinkling prevention (the boundary between the blue dotted line and the black dotted line) clearly separates the safe forming region from the wrinkling-prone unstable region. The internal pressure can continuously and stably increase and remain within the safe region under the optimized GLLF trajectory (with an initial gas volume fraction $k=0.6$). When the cross-sectional compression displacement reaches 1.0 mm, the actual support pressure corresponding to an initial pressure of 1.0 MPa reaches 2.96 MPa (marked with an asterisk), successfully achieving defect-free mold bonding. On the contrary, if the initial pressure

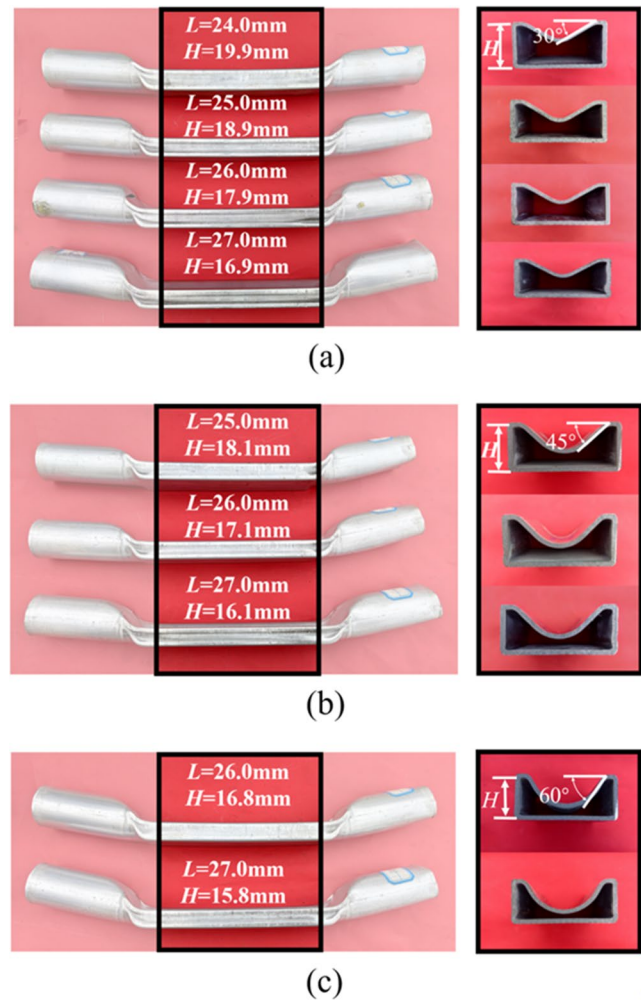


Fig. 6 Macroscopic experimental results and cross-sectional morphologies of tubes with different concave angles during the compression process: (a) 30° concave tube; (b) 45° concave tube; (c) 60° concave tube. (The slight deformation observed at the extremities was induced during the specimen removal process and has no impact on the analysis of the main deformation zone.)

is 0.8 MPa, at the same displacement, the actual pressure is only 2.41 MPa (marked with a cross symbol), and at this time, the tube wall immediately exhibits macroscopic wrinkling instability due to the inability to resist the planar longitudinal compression. As the compression deepens, the threshold for preventing wrinkling further increases. At a compression displacement of 2.0 mm, an initial pressure of 1.5 MPa yields a safe support pressure of 4.64 MPa, whereas an initial pressure of 1.0 MPa generates an internal pressure of only 4.03 MPa—insufficient to prevent unstable forming. At the ultimate compression displacement of 3.0 mm, maintaining die fitting requires an actual pressure as high as 6.88 MPa (corresponding to an initial pressure of 2.0 MPa); even at 5.83 MPa, wrinkling still occurs. The observed exponential hardening characteristic is consistent with the theoretical prediction of the self-pressurization mechanism

(Eq. 9), which states that the critical support pressure must rise sharply to resist the increase in stress during the subsequent deformation stage.

As the concave angle of the tube increases, the depth of the mold penetration into the local area deepens rapidly, leading to significantly intensifies the deformation. Consequently, the forming pressure curve of the 45° concave tube shifts markedly upward overall, as shown in Fig. 7(c). This is primarily attributed to the higher forming pressure required to achieve the 45° inclination of the concave feature while suppressing wrinkling defects (with an initial gas volume fraction $k=0.7$). The forming diagram in Fig. 7(d) visually shows that the increase in the critical pressure boundary position leads to a severe compression of the safe area and significantly increases the rigidity requirements for maintaining the stability of the side wall. At a compression displacement of 2.0 mm, the safe support pressure required to prevent tube instability rises to 7.28 MPa. Even if the internal pressure slightly decreases here (for example, to 6.23 MPa), it will cause the deformation trajectory to quickly cross the critical boundary and enter the wrinkling area, producing obvious geometric defects.

The 60° bent tube exhibits the most severe tendency of wrinkling instability among these three sets of tests. As shown in the pressure evolution curve in Fig. 7(e), its self-pressurization effect is extremely steep in the later stage (with an initial gas volume fraction $k=0.7$). At a compressive displacement of only 1.0 mm in the early stage of cross-sectional deformation (Fig. 7(f)), the safety loading boundary required to maintain the die fitting and prevent wrinkling has already reached 7.66 MPa. If internal pressure supplementation is insufficient—even a slight drop to 6.66 MPa—irreversible local wrinkling will be triggered. Moreover, the special deep V-shaped geometric feature causes the internal fluid volume to decrease sharply during compression, which strictly limits the ultimate compression displacement of the 60° concave tube to about 1.4 mm (at which point the critical supporting pressure surges to 9.51 MPa). This extremely limited compressible space is far lower than that of the 30° concave tube (greater than 3.0 mm), highlighting the extreme limits of process control for large-angle concave tubes.

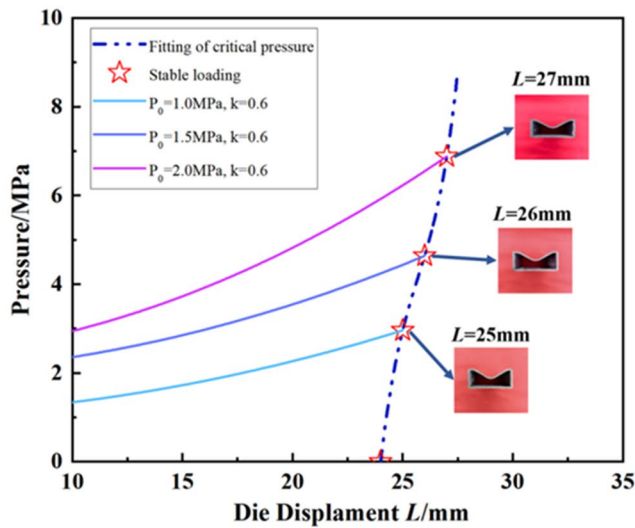
A comprehensive comparative analysis of the systematic data presented in the above figures reveals the core mechanical mechanism underlying the GLLF process: generally, the sharper the concave angle feature, the larger the local instability bending moment generated during in-plane compression. Consequently, the forming process must rely on an exponentially rising internal fluid pressure to effectively resist sidewall buckling. The GLLF process utilizes the non-linear self-pressurizing effect exhibited by the gas-liquid mixed medium in the later stages of forced compression,

and perfectly and dynamically matches this extremely stringent safe mechanical boundary requirement.

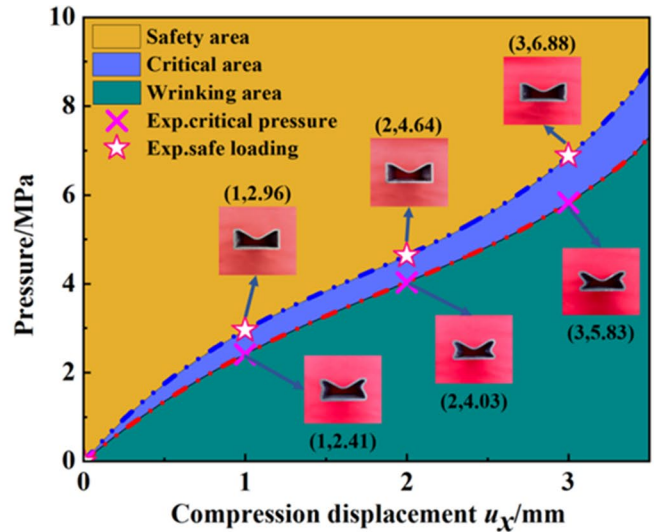
5.2 Analysis of deformation mechanism during the forming process

This study systematically analyzed the mechanical mechanisms throughout the entire forming process by combining the finite element circumferential stress (S22) distribution shown in Fig. 8, to further clarify the physical essence of tube shape evolution and wall thickness distribution. Figure 8(a) presents the established three-dimensional finite element model, which provides a basis for extracting the stress-strain state under complex mold constraints. As shown in Fig. 8(b), a complete circumferential path is extracted along the outer surface of the tube cross-section, starting from the bottom symmetrical center ($S=0$) and returning to the same position ($S=1$). Five specified key geometric feature points facilitate detailed quantitative analysis: the bottom convex corner (A, E), the top convex corner (B, D), and the central concave corner (C). Based on the stress evolution pattern, the entire forming process can be clearly divided into two stages: a “cross-sectional forming stage” dominated primarily by local bending, and a “cross-sectional compression stage” governed mainly by macroscopic compressive stress.

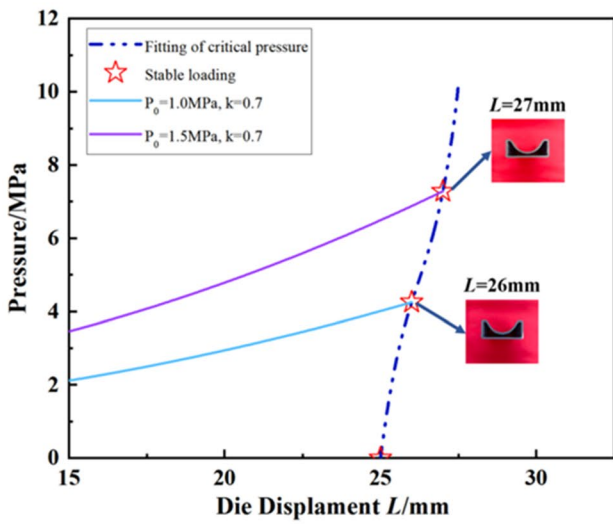
Figure 8(c) and (d) show the stress evolution and deformation characteristics of the 30° concave tube. As shown in the black and red curves in Fig. 8(d), the external surface stress fluctuates violently above and below the zero line during the early and middle stages of forming ($L=6, 15$ mm), reflecting a typical bending-dominated state. The central concave area (point C) exhibits an outward drawing tendency due to the penetration of the V-shaped upper mold, and it presents a moderate tensile stress. When the tube enters the final severe compression stage ($L=27$ mm) (as shown by the blue curve in Fig. 8(d) and the corresponding isosurfaces in Fig. 8(c)), the stress state undergoes a fundamental reversal. The stress level across the entire cross-section significantly sinks into the negative (compressive) regime. The side straight-edge segments (A-B and D-E segments) become the primary load transfer path, bearing extremely strong in-plane compressive stress. Notably, the peak compressive stress at the bottom convex corners A and E reaches approximately -150 MPa. In the 30° concave tube, the relatively gentle geometric transition enables the material to flow smoothly toward these corner areas, achieving effective “compression accumulation” and ensuring the tight adhesion of the mold. The mechanical response becomes more pronounced for the 45° concave tube in Fig. 8(e) and (f) due to the deeper concave feature. In the initial shaping stage ($L=6$ mm), the sharp penetration of the mold at the central concave corner (point C) causes a



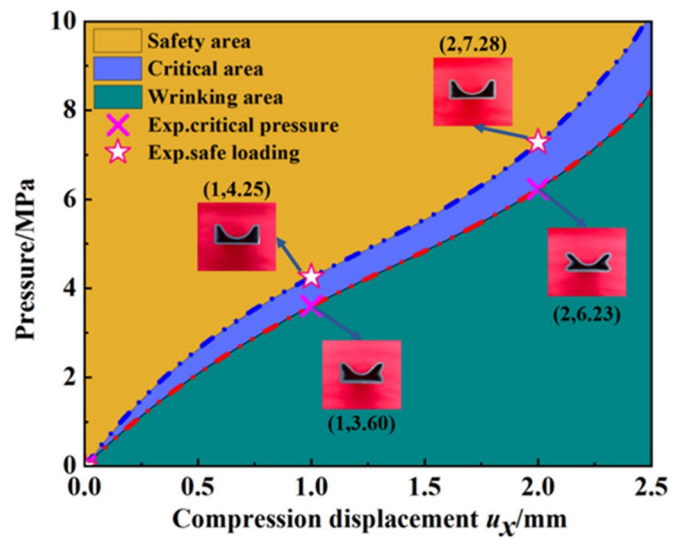
(a)



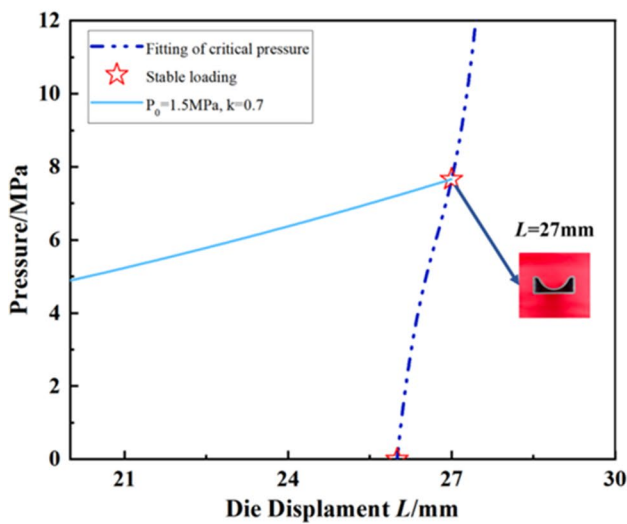
(b)



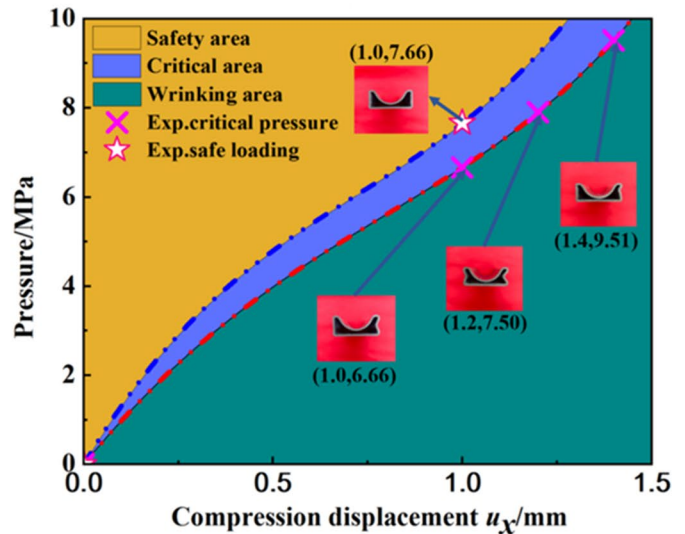
(c)



(d)



(e)



(f)

Fig. 7 Forming pressure-displacement curves (left column) and forming process windows determined under optimized working conditions (right column) for tubes with different concave angles: (a, b) 30° concave tube; (c, d) 45° concave tube; (e, f) 60° concave tube. (The safety boundaries, critical wrinkling boundaries, and experimental verification scatter points in the process windows are all obtained based on the optimized initial gas-liquid volume fraction, where $k=0.6$ for the 30° tube, and $k=0.7$ for the 45° and 60° tubes.)

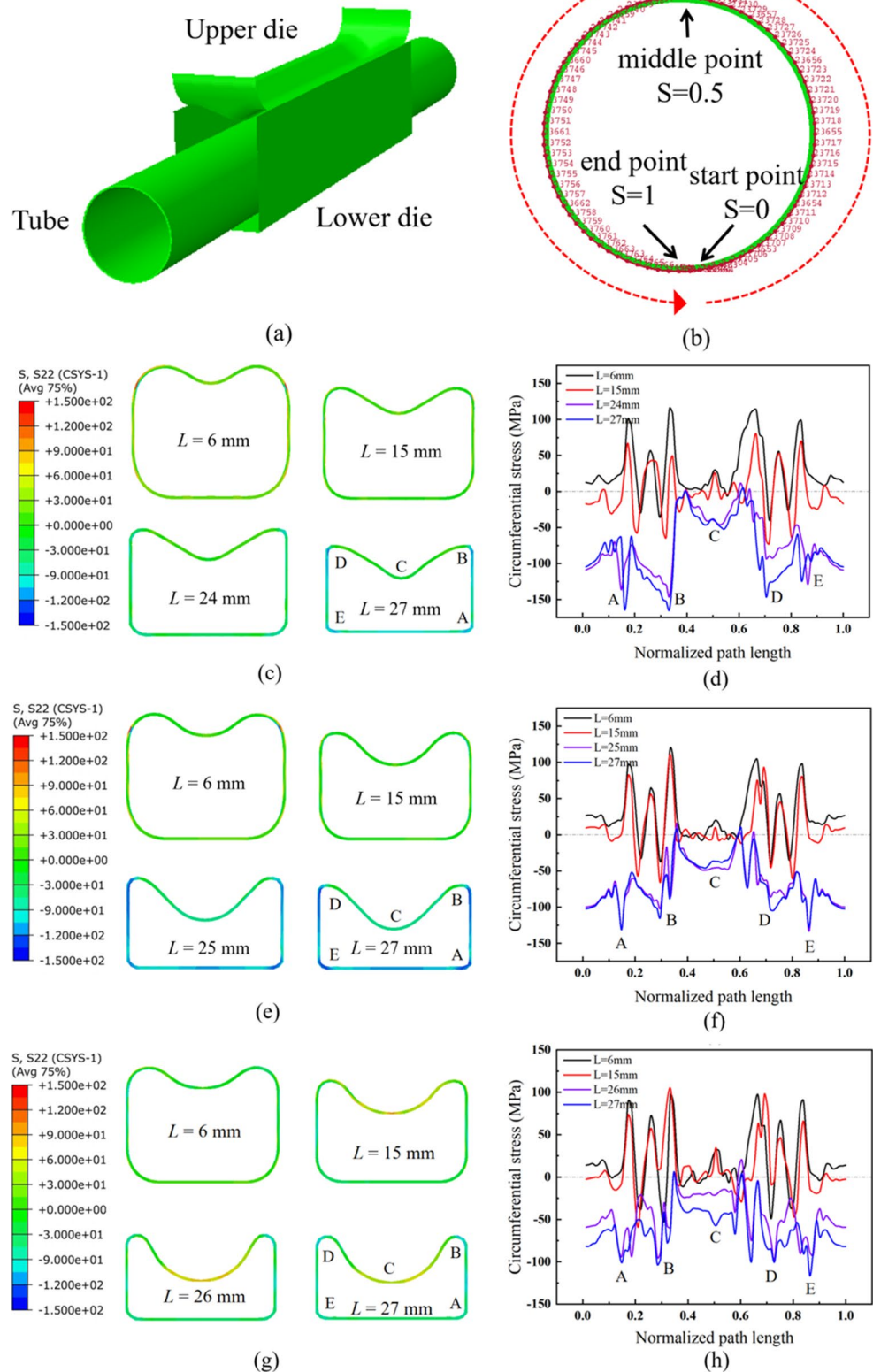
sudden increase in tensile stress, which is clearly visible as a prominent peak in Fig. 8 (f) and a local red area in the contour of Fig. 8(e). When the process transitions to the cross-sectional compression stage at $L=27$ mm, the blue curve in Fig. 8(f) shows a more extensive compressive stress distribution compared to the 30° concave, and the compressive stress at the top convex corners (B, D) increases significantly. This indicates that the 45° concave feature increases the resistance to lateral material flow and requires a higher “self-boosting” effect from the internal fluid to stabilize the straight-edge area under such high compressive loads, thereby preventing the occurrence of macroscopic bending. Figure 8(g) and (h) of the 60° concave tube exhibit the most extreme mechanical performance among the three cases. The deep V-shaped mold generates a powerful “mechanical tensile” effect in the initial stage ($L=6$ mm), producing in a tensile stress peak exceeding 100 MPa at the central concave area (point C) in Fig. 8(h). Although the global compression at $L=27$ mm eventually forces the stress curve to move downward, the sharp geometric shape of the 60° concave angle generates a “locking effect” that severely impedes the transverse material compensation. As shown by the blue curve in Fig. 8(h), the compressive stress level at point C (approximately -50 MPa) is significantly lower than that of the adjacent straight sections. This indicates that the later compression cannot fully compensate for the thinning induced in the early stage. This mechanical process provides a fundamental explanation for the maximum thinning of 5.5% observed in the central region of the 60° concave tube. The severe initial stretching and the limited material compensation in the later stage jointly determine the final forming limit. In summary, the GLLF process exploits the sharp increase in internal pressure during the compression stage to establish a strong normal supporting force in the straight sections. This mechanism ensures that the tube can avoid wrinkling defects even when subjected to extremely high circumferential compressive stress, as evidenced by the stress states at points A, B, D, and E in Figs. 8(d), 8(f), and 8(h). The synergy between die-driven compression and adaptive internal-pressure support ultimately enables high-precision manufacturing of complex concave components through a composite “tension–bending–compression” mechanical mechanism.

5.3 Analysis of tube wall thickness distribution during the forming process

Figures 9 and 10 present the ultrasonic thickness measurement data and the ABAQUS shell thickness (STH) simulation cloud maps, respectively. Together, they reveal the wall thickness distribution characteristics and the underlying material flow laws of tubes with different concave angles (30°, 45°, 60°) during the GLLF process. The full-field wall thickness distribution characteristics obtained from the simulation show the high consistency with the experimental measurement results, which confirms the unique deformation mechanism of the GLLF process—one that is fundamentally distinct from conventional high-pressure tube hydroforming. Specifically, the material flow is characterized by active “centripetal accumulation” rather than passive overall tensile thinning, driven by the dominant action of mechanical die compression. Moreover, the severe deformation of the tube cross-section is mainly concentrated in the straight-edge sections and the concave corner transition zones, which evolve into the primary “material accumulation zones” during the forming process. For the tubes with an initial wall thickness of $t=1.5$ mm, the wall thickness in the flat top and bottom regions remains basically at the initial level (thickness change rate less than 5%) at the early and middle compression stages due to the minimal equivalent plastic strain. However, the material is forced to migrate circumferentially from the gentle top and bottom regions to the straight-edge sections and corners because of the macroscopic circumferential compressive stress. Subsequently, the cross-regional flow of the material is hindered under the combined action of the strong geometric constraints imposed by the concave profile and the local multiaxial compressive stress state, resulting in significant compressive accumulation. The severity of this tube wall thickening effect is highly sensitive to the concave angle.

As shown in Fig. 9(a), for the tube with a 30° concave angle, the gentle geometric transition results in the lower lateral feeding resistance. The simulation cloud contour in Fig. 10 shows that significant material accumulation occurs in the corner region after entering the critical point and transitioning to the section compression stage, and with the maximum wall thickness reaching 1.650 mm. This is consistent with the peak wall thickness measured by ultrasound in Fig. 9 (approximately 1.685 mm, a local thickening rate of 12.3%). However, as the concave angle increases, the steeper geometric profile considerably increases the frictional resistance and deformation resistance at lateral material flow and effectively suppresses the excessive migration of material toward the top corners. Therefore, the thickening

Fig. 8 Finite element analysis of circumferential stress (S22) evolution during the Gas-Liquid Low-pressure Forming process for thin-walled tubes with different concave angles: **(a)** 3D simulation model; **(b)** definition of the stress extraction path along the outer surface of the tube cross-section; **(c-d)** stress contours and corresponding 2D path curves for the 30° concave tube (regions A-E correspond to key geometric features); **(e-f)** for the 45° concave tube; **(g-h)** for the 60° concave tube



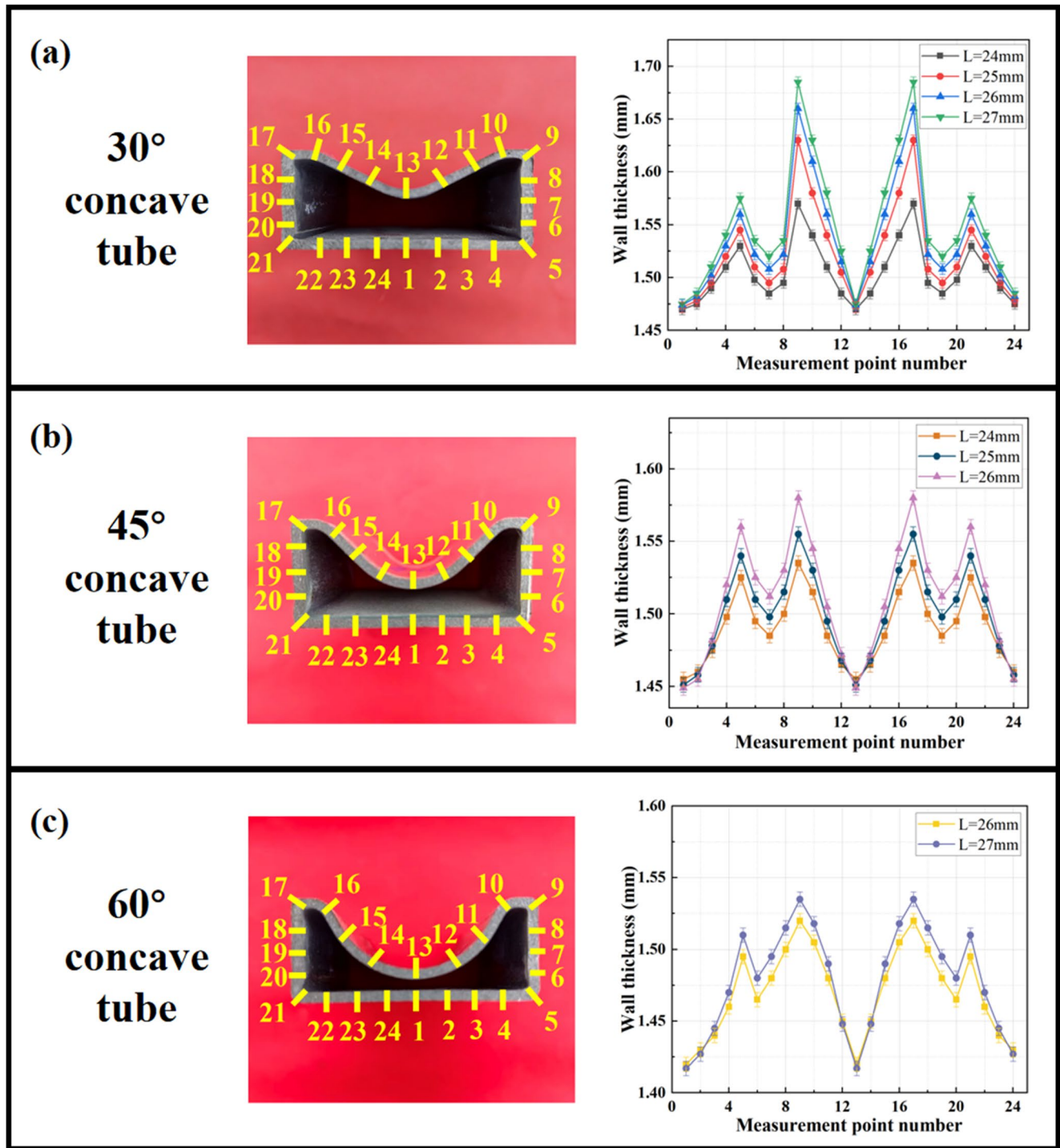


Fig. 9 Ultrasonic measurement results of wall thickness distribution along cross-sectional feature points under different punch strokes: (a) 30° concave tube; (b) 45° concave tube; (c) 60° concave tube. (Note:

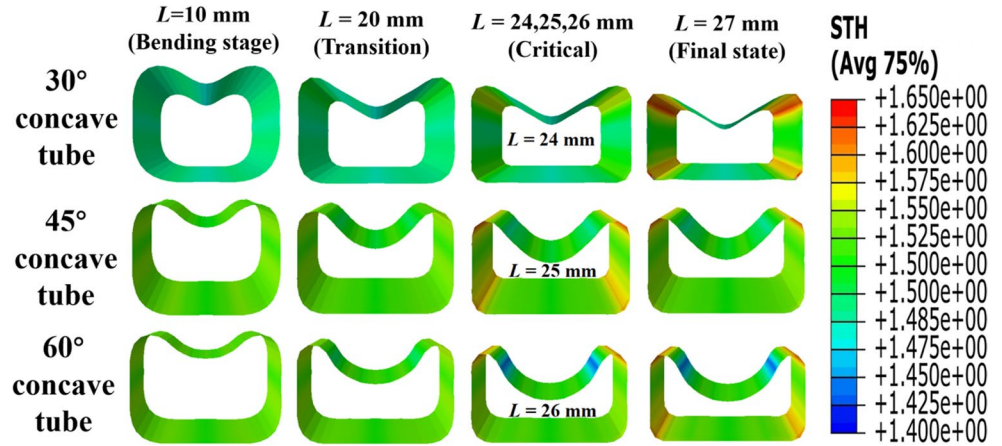
The insets show the distribution of ultrasonic thickness measurement sampling points.)

phenomenon in the corner regions is markedly alleviated in the 45° and 60° concave tubes. In Fig. 9(b) and (c), ultrasonic measurements show the maximum wall thicknesses dropping to approximately 1.580 mm and 1.535 mm, respectively. The extreme value distribution in the simulation

cloud contours of the concave tubes in Fig. 10 simultaneously verifies this attenuation trend.

In stark contrast to the multi-axial compression thickening mechanism in the corner regions, the central areas of the top and bottom depressions of the tube primarily face

Fig. 10 Finite element simulation contour maps of wall thickness (STH) distribution evolution for tubes with different concave angles (30°, 45°, and 60°) under typical forming stages (bending stage, transition, critical stage, and final state)



localized thinning dominated by bidirectional tensile stress. As shown in Fig. 9(a), for the 30° concave tube, the wall thickness in the central region remains essentially around 1.470 mm throughout the forming cycle, with no significant strain localization due to the smooth overall material flow. As the concave angle increases to 45° and 60°, the overall deformation resistance of the tube rises substantially. The steeper geometric features cause the tensile strain to be highly concentrated along the symmetry center plane during the die downward displacement. The wall thickness measurement data in Fig. 9(b) and (c) explicitly indicate that the minimum wall thickness in the center of the 45° concave tube drops to 1.449 mm, whereas the central region of the 60° concave tube exhibits the most severe thinning, with the wall thickness dropping sharply to 1.417 mm (a thinning ratio of 5.5%). The STH contour maps in Fig. 10 also show distinct cyan low-thickness bands in the central region in the final state, further confirming the amplifying effect of high-angle concave features on local tensile stress concentration.

It is evident that the forming quality and dominant defect modes of concave tubes under the GLLF process are strictly controlled by the concave angle. At gentle concave angles (e.g., 30°), material flow resistance is low, and the primary forming limitation lies in local wrinkling or out-of-tolerance corner wall thickening caused by excessive centripetal material accumulation. On the contrary, at steeper concave angles (e.g., 60°), deformation resistance increases substantially, and the geometric constraints impede effective material compensation. Therefore, the dominant failure mode of the tube shifts to excessive tensile thinning and even rupture in the central region, driven by severe stress concentration. This unique wall thickness evolution mechanism not only avoids the thinning and cracking risks that occur in traditional processes, but also provides a theoretical basis for enhancing the local stiffness of the tube by adjusting the concave corner angle.

It is worth noting that the observed maximum thinning rate of 5.5% for the 60° concave tube lies well below the

safety limit of the 6061 aluminum alloy, indicating excellent formability under the current Gas-Liquid Low-Pressure Forming (GLLF) loading path. However, numerical sensitivity trends suggest that if the concave angle were increased further (e.g., >60°), the localized geometric discontinuity would induce severe tensile stress concentration around the transition corners. This would restrict fluid-driven material flowing toward the center and shift the dominant strain path back toward localized tension, thereby sharply elevating the thinning rate toward the necking threshold. Hence, while 5.5% ensures a robust safety margin for 60° tubes, it marks the transition regime beyond which steeper angles would necessitate careful optimization of the internal pressurization to prevent defects.

5.4 Analysis of the forming accuracy

To comprehensively evaluate the dimensional control capability of the GLLF process for manufacturing tubes with complex cross-sections, this study utilized 3D optical scanning technology to quantitatively assess the geometric accuracy of the formed tubes. As shown in Fig. 11(a), 30°, 45°, and 60° concave tubes are fabricated under their respective optimized process parameters, exhibiting macroscopically smooth surfaces and no obvious instability defects. The most representative deep concave feature in the middle region is selected as the test object, as enclosed by the red dashed box. As shown in Fig. 11(b), high-contrast optical tracking markers are evenly arranged on the surface of the intercepted feature samples, and high-precision point cloud data are acquired using a 3D optical scanner. The actual scanned profiles are perfectly aligned and fitted with the ideal theoretical CAD models to generate global 3D scanning deviation contour maps, as shown in Fig. 11(c) (the green color in the contours represents the ideal forming region tending toward zero deviation).

Overall, the GLLF process demonstrates excellent forming accuracy. Despite the varying concave angles and

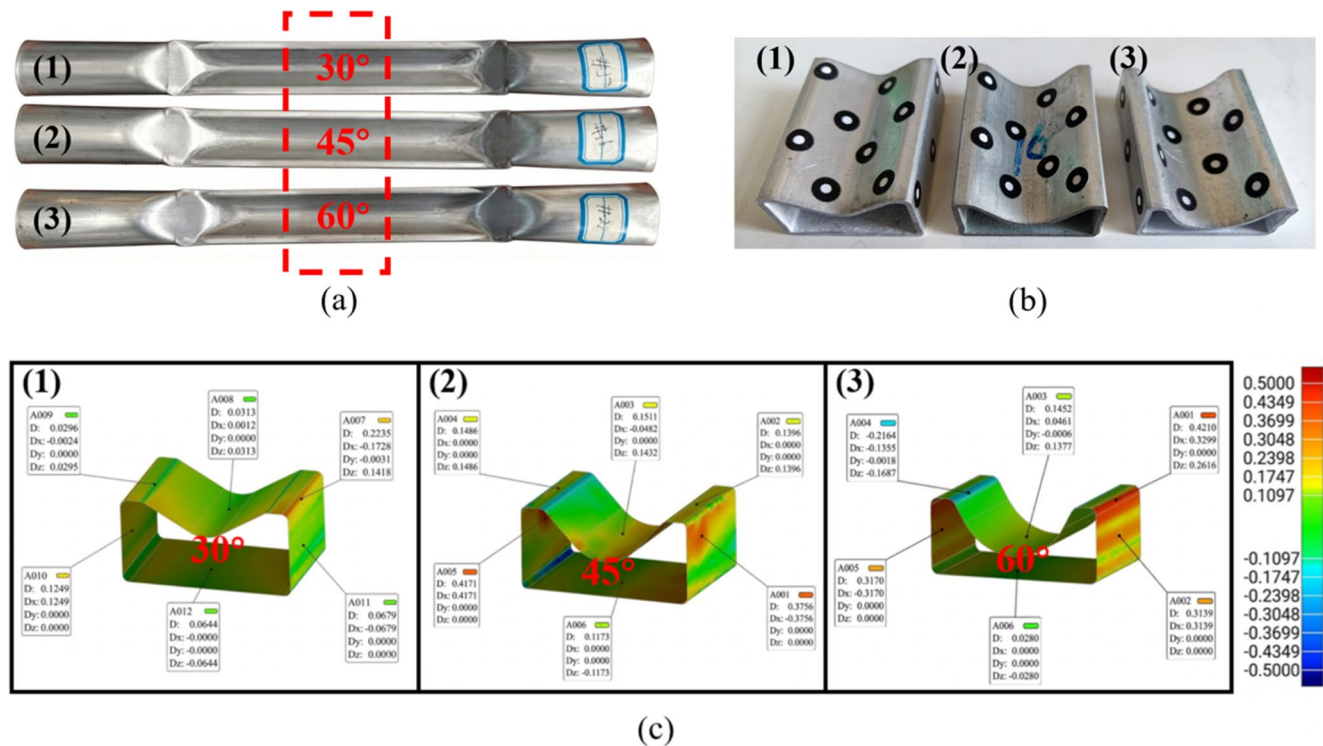


Fig. 11 Experimental evaluation of forming accuracy and 3D scanning deviation for tubes with different concave angles: (a) macroscopic view of the formed tubes and the selected scanning region (red dashed

box); (b) sectioned samples with optical tracking markers for 3D scanning; (c) 3D scanning deviation contours corresponding to (1)-30°, (2)-45°, and (3)-60° concave tubes

geometric complexities of the tubes, the global maximum profile deviations are all strictly controlled within 0.5 mm. Specifically, as shown in Fig. 11(c-1), for the 30° concave tube formed under an initial pressure of 1.0 MPa and an initial gas fraction $k=0.6$, the internal mixed medium adaptively builds up to a final pressure of 2.96 MPa as the die descends. Sufficient internal pressure support enables that most of the tube surface appear green, with a maximum overall geometric deviation of only approximately 0.22 mm (e.g., at test point A007). As the concave angle increases, both the cross-sectional forming difficulty and the material deformation resistance rise significantly, yet the GLLF process still maintains extremely high dimensional accuracy. As shown in Fig. 11(c-2), the final forming pressure of the 45° concave tube adaptively climbs to 7.28 MPa under the conditions of an initial pressure of 1.5 MPa and initial gas fraction $k=0.7$, and the maximum profile deviation at its sidewalls and transition corners is approximately 0.42 mm (e.g., test point A005). As shown in Fig. 11(c-3), the 60° concave tube possesses the steepest and most complex deep concave features under the same initial parameters (initial pressure of 1.5 MPa and gas fraction $k=0.7$), which drives its final forming support pressure further up to 7.66 MPa. However, its maximum surface deviation remains stable at around 0.42 mm (e.g., test point A001). The comprehensive comparison results indicate that the nonlinear

self-pressurizing effect generated by the gas-liquid hybrid fluid as the compression volume decreases can well accommodate the rigid dynamic requirements for die adhesion pressure across features of different angles and depths. As a result, the high-precision manufacturing of complex concave thin-walled metal tubes is successfully achieved without increasing the complexity of die control.

6 Conclusion

To overcome the engineering challenges associated with traditional high-pressure tube hydroforming and low-pressure forming, this paper proposes a Gas-Liquid Low-pressure Forming (GLLF) process for fabricating thin-walled metal tubes with complex concave cross-sections. The main research conclusions are as follows:

- (1) A theoretical model for the compression of gas-liquid mixed media was established. The GLLF process achieved a nonlinear adaptive pressure response characterized by an “initially gradual, then steep” profile, through the dynamic adjusting of the gas volume fraction during compression and proper matching of the initial pressure, thereby effectively satisfying the anti-instability requirements of the late forming stage.

- (2) The critical supporting pressure boundaries were accurately defined. It also indicated that the steeper the concave angle, the exponentially increasing internal pressure was required to prevent the instability of the straight-edged structure (The minimal internal pressure required increases from 1.2 MPa for 30° tubes to 1.6 MPa for 45° tubes and 2.0 MPa for 60° tubes). Although the steep geometric shape of the tube had extremely strong resistance to deformation, this process still strictly controlled the maximum three-dimensional contour deviation of the convex and concave tube to be within 0.5 mm. The maximum contour deviations for 30°, 45° and 60° concave tubes were 0.22 mm, 0.42 mm and 0.42 mm respectively.
- (3) This process transformed the material deformation mode from “tensile thinning” to “compressive accumulation”. For the 30° concave tube with a gentle angle, the smooth backflow of the material led to a local thickening of 12.3% at the transition corners. In contrast, the geometric constraints of the 60° tube with a steeper angle increased the transverse flow resistance, resulting in a maximum thinning of approximately 5.5% in the central part.
- (4) The principle of defect suppression under the “tension-bending-compression” composite mechanism was clarified. The straight edge section was subjected to extremely high in-plane compressive stress. For a 30° concave tube, the peak compressive stress reached approximately 150 MPa. The adaptive self-pressurization effect of the GLLF process provided a strong normal supporting force to effectively eliminate the bending wrinkles.

From an industrial application perspective, the proposed GLLF technology provides a highly efficient and cost-effective manufacturing solution for complex concave thin-walled tubes, as demonstrated by its successful deployment in producing premium aluminum bicycle frame tubes (e.g. Giant Manufacturing Co.Ltd.). The implementation of durable conical sealing punches avoids frequent replacement of dynamic seals, while the gas-liquid self-pressurization system eliminates the need for expensive high-pressure compressed air source, thereby substantially reducing both equipment investment and unit production cost. Furthermore, this process achieves excellent dimensional accuracy (profile deviation < 0.5 mm) and ultra-fast cycle times (< 15s/part) with high production consistency. Overall, GLLF offers a possibility for improving the manufacturing process of concave structural components in the fields of high-end bicycles, automobiles, and lightweight transportation.

Author contributions Zhijie Hu: Writing-review & editing, Writing-original draft, Software, Methodology, Investigation. Hang Yuan: Writing-review & editing, Supervision. Junzhuo Wang: Writing-re-

view & editing, Supervision. Lixiang Li: Writing-review & editing, Supervision. Yanli Lin: Resources, Supervision. Dongshen Xiang: Writing-review & editing, Supervision. Zhubin He: Supervision, Funding acquisition, Conceptualization.

Funding This work was supported by the Changjiang Scholar Program, Ministry of Education (No. T2024047). The authors wish to express their gratitude to the fund.

Data availability The data used to support the findings of this study are available from the corresponding author upon request.

Code availability Not applicable.

Declarations

Ethical approval Not applicable.

Consent to participate All authors agreed to participate in this research.

Consent for publication All authors have read and agreed to the published version of the manuscript.

Conflict of interest The authors declare no competing interests.

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